

JEE MAIN REVISION MASTER: ELECTROMAGNETIC INDUCTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 153 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Magnetic Flux & Faraday-Lenz Law	138 PYQs	Core Principle
Very High	Self Inductance & RL Transient Circuits	41 PYQs	Inductance & Circuits
Very High	Motional EMF (Translational & Rotational)	38 PYQs	Motional Induction
Very High	Mutual Inductance & Coupled Coils	21 PYQs	Coupling
Moderate	Induced Electric Field & Eddy Currents	4 PYQs	Field Theory
Moderate	Electromagnetic Induction - Core Application	6 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Magnetic Flux & Faraday-Lenz Law	$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos\theta$, $\mathcal{E} = -\frac{d\Phi_B}{dt} = -N \frac{d\Phi_B}{dt}$, $q = \frac{\Delta\Phi}{R}$
Self Inductance & RL Transient Circuits	$\mathcal{E} = -L \frac{dI}{dt}$, $L = \mu_0 n^2 A l$, $U = \frac{1}{2} L I^2$, $\tau = \frac{L}{R}$, $I(t) = \frac{V}{R} (1 - e^{-t/\tau})$
Motional EMF (Translational & Rotational)	$\mathcal{E} = B v L$, $I = \frac{B v L}{R}$, $F_{\text{mag}} = \frac{B^2 L^2 v}{R}$, $\mathcal{E}_{\text{rot}} = \frac{1}{2} B \omega L^2$
Mutual Inductance & Coupled Coils	$\mathcal{E}_2 = -M \frac{dI_1}{dt}$, $M = \mu_0 n_1 n_2 (\pi r_{\text{inner}}^2) l$, $M = k \sqrt{L_1 L_2}$ ($0 \leq k \leq 1$)
Induced Electric Field & Eddy Currents	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$, $E(r) = \frac{r}{2} \frac{dB}{dt}$; ($r \leq R$), $E(r) = \frac{R^2}{2r} \frac{dB}{dt}$; ($r > R$)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Magnetic Flux & Faraday-Lenz Law (Very High Yield — 138 PYQs)

Core Principle: Magnetic flux $\Phi = B \cdot A \cdot \cos(\theta)$. Induced EMF $e = -d\Phi/dt$. Lenz's law dictates that the induced current opposes the change in magnetic flux producing it.

■ **Caution / JEE Pitfall:** 1. Charge flown through circuit $\Delta q = \Delta \Phi / R$ is independent of the rate/time of flux change! 2. Angle θ is between B and the normal to the coil area, not the coil plane.

■ **Speed Trick:** When a coil of N turns rotates with angular velocity ω : $e = NBA\omega \sin(\omega t)$, $e_{\max} = NBA\omega$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A 30 cm long solenoid has 10 turns per cm and area of 5cm^2 . The current through the solenoid coil varies from 2 A to 4 A in 3.14 s. $T = \dots$ " [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "A circular current loop of radius R is placed inside square loop of side length L ($L \geq R$) such that they are co-planar and their centre..." [Ans: D]
- [2026 Evening Shift] Q#1 (MCQ): "Suppose a long solenoid of 100 cm length, radius 2 cm having 500 turns per unit length, carries a current $I = 10 \sin(\omega t)$ A, where $\omega = \dots$ " [Ans: N/A]

3.2 Self Inductance & RL Transient Circuits (Very High Yield — 41 PYQs)

Core Principle: Self-inductance $L = \Phi / I$, induced back EMF $e = -L(dI/dt)$. Solenoid $L = \mu_0 \cdot n^2 \cdot A \cdot l$. RL circuit current growth: $I(t) = I_0(1 - e^{-t/\tau})$, $\tau = L/R$.

■ **Caution / JEE Pitfall:** At $t = 0+$ (immediately after switch closing), inductor acts as an open circuit (infinite resistance). At $t \rightarrow \infty$ (steady state), inductor acts as an ideal zero-resistance short wire.

■ **Speed Trick:** Magnetic energy density in space: $u_B = B^2 / (2\mu_0)$. Inductor resists sudden changes in current, not voltage.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A 30 cm long solenoid has 10 turns per cm and area of 5cm^2 . The current through the solenoid coil varies from 2 A to 4 A in 3.14 s. $T = \dots$ " [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "Suppose a long solenoid of 100 cm length, radius 2 cm having 500 turns per unit length, carries a current $I = 10 \sin(\omega t)$ A, where $\omega = \dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (Numerical): "Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω . The energy stored in the coil is..." [Ans: 20]

3.3 Motional EMF (Translational & Rotational) (Very High Yield — 38 PYQs)

Core Principle: EMF induced in a conductor of length L moving with velocity v perpendicular to B : $e = B \cdot v \cdot L$. For a rod rotating about one end: $e = (1/2) \cdot B \cdot \omega \cdot L^2$.

■ **Caution / JEE Pitfall:** 1. Mechanical power delivered by external agent $F \cdot v$ equals electrical power dissipated $I^2 R$. 2. For an irregularly shaped wire moving in B , replace it with the straight displacement vector connecting its endpoints.

■ **Speed Trick:** Direction of motional EMF: find direction of $(v \times B)$. Positive charges accumulate at the end pointed to by $(v \times B)$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Three identical coils C_1, C_2 and C_3 are closely placed such that they share a common axis. C_2 is exactly midway. C_1 carries current I into the page. The induced current in C_3 is..." [Ans: B]
- [2026 Morning Shift] Q#2 (MCQ): "A 20 m long uniform copper wire held horizontally is allowed to fall under the gravity ($g = 10\text{m/s}^2$) through a uniform horizontal magnetic field of 0.1T directed into the page. The induced EMF in the wire is..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (MCQ): "A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10T . If the rod is pulled to the right with a constant velocity v , the induced EMF in the rod is..." [Ans: B]

3.4 Mutual Inductance & Coupled Coils (Very High Yield — 21 PYQs)

Core Principle: Flux in secondary due to current in primary $\Phi_{21} = M I_1$. Induced EMF $e_2 = -M(dI_1/dt)$. Coaxial solenoids $M = \mu_0 n_1 n_2 \pi r_1^2 l$.

■ **Caution / JEE Pitfall:** In coaxial solenoids, always use the area of the INNER coil (r_{inner}), because magnetic field of outer coil only exists over the area of inner coil.

■ **Speed Trick:** Reciprocity theorem holds: $M_{12} = M_{21}$ regardless of differing geometry or number of turns.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A circular current loop of radius R is placed inside square loop of side length L ($L \geq R$) such that they are co-planar and their cente..." [Ans: D]
- [2026 Morning Shift] Q#2 (Numerical): "In the given circuit below inductance values of L_1, L_2 and L_3 are same. The magnetic energy stored in the entire circuit is $(U \dots)$ " [Ans: 6]
- [2025 Morning Shift] Q#1 (MCQ): "Consider I_1 and I_2 are the currents flowing simultaneously in two nearby coils 1 & 2, respectively. If L_1 = self inductance of coil 1, $M \dots$ " [Ans: N/A]

3.5 Induced Electric Field & Eddy Currents (Moderate Yield — 4 PYQs)

Core Principle: Time-varying magnetic field produces non-conservative circular electric field:oint $E \cdot dl = -d\Phi/dt$. $E = (r/2) \cdot (dB/dt)$ for $r \leq R$, $E = (R^2 / (2r)) \cdot (dB/dt)$ for $r > R$.

■ **Caution / JEE Pitfall:** Induced electric field is non-electrostatic: field lines form closed loops and work done along a closed path is non-zero!

■ **Speed Trick:** Eddy currents are minimized by laminating metallic cores with insulation (used in transformers and motors).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (Numerical): "Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω . The en..." [Ans: 20]
- [2023 Morning Shift] Q#1 (MCQ): "Certain galvanometers have a fixed core made of non magnetic metallic material. The function of this metallic material is..." [Ans: N/A]
- [2023 Evening Shift] Q#2 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A: A bar magnet dropped t..." [Ans: B]

3.6 Electromagnetic Induction - Core Application (Moderate Yield — 6 PYQs)

Core Principle: Key problem pattern in Electromagnetic Induction.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2023 Morning Shift] Q#2 (Numerical): "As per the given figure, if $(dI/dt) = -1A/s$ then the value of V_{AB} at this instant will be _____ V" [Ans: 30]
- [2022 Evening Shift] Q#1 (Numerical): "For the given circuit the current through battery of 6 V just after closing the switch 'S' will be _____ A...." [Ans: 1]
- [2021 Evening Shift] Q#1 (Numerical): "If the maximum value of accelerating potential provided by a radio frequency oscillator is 12 kV. The number of revolution made by a proton ..." [Ans: 543]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A 30 cm long solenoid has 10 turns per cm and area of 5cm^2 . The current through the solenoid coil varies from 2 A to 4 A in 3.14 s. The e.m.f. induced in the coil is $\alpha \times 10^{-5}\text{V}$. The value α is ____.

- (A) 60
- (B) 12 [Correct]
- (C) 120
- (D) 34

Correct Answer: B

Solution Analysis: According to Faraday's Law, a change in the magnetic flux (Φ) through a coil induces an electromotive force (e.m.f.) in that coil. The magnitude of the induced e.m.f. (ϵ) is given by: $\epsilon = N(d\Phi / dt)$ Where N is the total number of turns. For a solenoid, the magnetic flux Φ through each turn is $\Phi = B \cdot A$, where B is the magnetic field inside the solenoid and A is the cross-sectional area. The magnetic field B inside an ideal solenoid is: $B = \mu_0 n I$ Where μ_0 is the permeability of free space ($4\pi \times 10^{-7}\text{T} \cdot (\text{m} / \text{A})$), n

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Morning Shift (Q#2)

In the given circuit below inductance values of L_1, L_2 and L_3 are same. The magnetic energy stored in the entire circuit is ($U(t)$) and that stored in the L_2 inductor is ($U(I)$). $U(t)/U(I)$ is ____ . (Ignore the mutual inductance if any)

Correct Answer: 6

Solution Analysis: The magnetic energy (U) stored in an inductor of inductance L carrying a current I is given by the formula: $U = (1/2)LI^2$ In the given circuit L_1 is in series with a parallel combination of L_2 and L_3 . Let the total current entering the circuit be I. Then the current through L_1 is $I_1 = I$. Since L_2 and L_3 are in parallel and have the same inductance ($L_2 = L_3 = L$), the current I will distribute equally between them. So, the current through L_2 is $I_2 = (I/2)$ and current through L_3 is $I_3 = (I/2)$. Using the energy formula